A Preliminary Result of Completely Independent Spanning Trees on Chordal Rings*

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Abstract

Let $T_1, T_2, \ldots, T_k$ be spanning trees in a graph G. If, for any two vertices u, v of G, the paths joining u and v on the k trees are mutually vertex-disjoint, then $T_1, T_2, \ldots, T_k$ are called completely independent spanning trees (CISTs for short) in G. The construction of CISTs can be applied in fault-tolerant broadcasting and secure message distribution on interconnection networks. Hasunuma first introduced the concept of CISTs and conjectured that there are k CISTs in any $2k$-connected graph. Although this conjecture was, unfortunately, disproved by Péterfalvi recently. In this paper, we show that there are two CISTs in 4-regular chordal rings $CR(N, d)$ with $N = k(d-1)+j$ under the conditions that $k \geqslant 4$ is even and $0 \leqslant j \leqslant 4$. Moreover, the diameter of each constructed CIST is derived.

Keyword: independent spanning trees; completely independent spanning trees; chordal rings.

1. Introduction

Let G be a simple undirected graph with vertex set $V(G)$ and edge set $E(G)$. A graph G is *k-connected* if $|V(G)| > k$ and $G - F$ is connected for every set $F \subset V(G)$ with $|F| < k$, where $G - F$ denotes the graph of G that removes all vertices of F. For $x, y \in V(G)$, two paths joining x and y are said to be *vertex-disjoint* (or *internally vertex-disjoint*) if they have no common vertex except x and y, and *edge-disjoint* if they share no common edge. A *tree* is an acyclic and connected graph. A *leaf* in a tree is a vertex with degree 1. A *rooted tree* is a tree with a distinguished vertex called the root. A tree T in a graph G is a *spanning tree* if $V(T) = V(G)$.

A set of k spanning trees of G rooted at a vertex r is said to be *independent* if the paths from any vertex $v(\neq r)$ to r on the k trees are mutually vertex-disjoint. A set of k spanning trees of G is said to be *completely independent* if for any two vertices $x, y \in V(G)$, the paths joining x and y on the k trees are mutually vertex-disjoint. Independent spanning trees (respectively, completely independent spanning trees) are sometimes referred to as ISTs (respectively, CISTs) for abbreviation. Also, we say that two spanning trees T_1 and T_2 are *edge-disjoint* if $E(T_1) \cap E(T_2) = \emptyset$.

Constructing ISTs or CISTs has many applications on interconnection networks such as fault-tolerant broadcasting and secure message distribution [2,27,33]. Zehavi and Itai [40] conjectured that any k-connected graph G has k ISTs rooted at an arbitrary vertex r of G. Since the conjecture remains an open problem for $k \geqslant 5$ (see [9,10,16]), a lot of endeavors have been devoted to this research. In particular, the conjecture has been confirmed for many special classes of graphs related to interconnection networks [5–8, 17–21, 27–39].

By contrast, Hasunuma [13] conjectured that there are k CISTs in any $2k$-connected graph and showed the NP-completeness for determining the existence of two CISTs in an arbitrary graph G. So far the study related to CISTs has received less attention except for [13–15, 24]. Hasunuma showed that there are k CISTs in the underlying graph of any k-connected line digraph $L(G)$ [13], there are two CISTs in any 4-connected maximal planar graph [14], and there are two CISTs in the Cartesian product of any 2-connected graphs [15]. Pai et al. [24] showed that there are $\lfloor \frac{n}{2} \rfloor$ CISTs in a complete graph with $n(\geqslant 4)$ vertices, there are $\lfloor \frac{n}{2} \rfloor$ CISTs in a complete bipartite graph $K_{m,n}$ with

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$m \geqslant n \geqslant 4$, and there are $\lfloor \frac{n_1+n_2}{2} \rfloor$ CISTs in a complete tripartite graph K_{n_3,n_2,n_1} with $n_3 \geqslant n_2 \geqslant n_1$ and $n_1 + n_2 \geqslant 4$. Recently, Péterfalvi [26] gave counterexamples to disprove Hasunuma's conjecture. He showed that there exists a k-connected graph which does not contain two CISTs for each $k \geqslant 2$.

In this paper, we try to carry on a preliminary research of CISTs on a family of interconnection networks called chordal rings. The remaining part of this paper is organized as follows. Section 2 gives the definition of chordal rings and provides an important property of CISTs. Section 3 presents our constructing schemes of CISTs for chordal rings under some restricted conditions. Finally, a concluding remark is given in the last section.

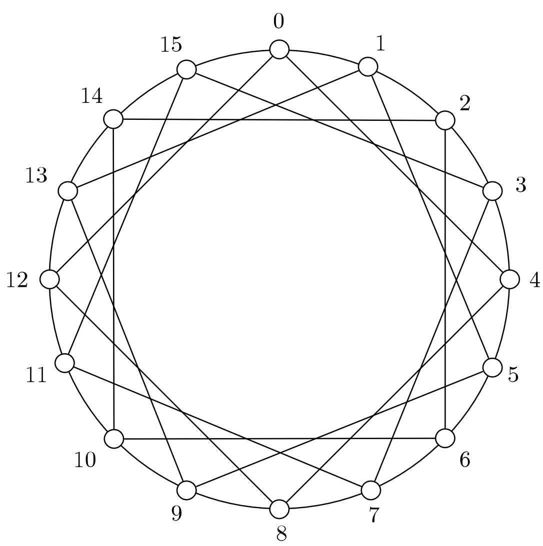

Figure 1: Chordal ring $CR(16, 4)$.

2. Preliminaries

Chordal rings (also called *distributed loop networks*) are a variation of ring networks. A chordal ring $CR(N, d)$ is a graph with vertex set $V = \{0, 1, 2, \ldots, N-1\}$ and edge set $E = \{(u, v) \mid u, v \in V$ and $(u - v) \equiv 1$ or $d \pmod N\}$. Obviously, $|E| = 2n$. For notational convenience, all arithmetics applied to a chordal ring are taken modulo N. To ensure that every vertex has degree 4, we assume $N > 4$ and $d < N/2$. Hence, the four edges $(u, u-1), (u, u+1), (u, u-d), (u, u+d)$ are said to be incident to a vertex $u \in V$ with *hop* $-1, +1, -d$ and $+d$, respectively. Also, $(u, u+1)$ and $(u, u+d)$ are called *forward edges* of u, and $(u, u-1)$ and $(u, u-d)$ are called *backward edges* of u. For example, a chordal ring $CR(16, 4)$ is shown in Figure 1. Chordal rings were introduced to enhance the reliability and fault-tolerance of ring networks [1, 3, 11, 12]. Previous researches for investigating algorithmic properties of chordal rings can refer to [4, 22, 23, 25, 41]. In particular, Iwasaki et al. [17] proposed a linear time algorithm for constructing four ISTs on a chordal ring. At a later time, Yang et al. [36] make an improvement on the construction by reducing the heights of ISTs.

Hasunuma [13] provided the following characterization for CISTs.

Theorem 1. [13] *$T_1, T_2, \ldots, T_k$ are CISTs in a graph G if and only if they are edge-disjoint and for any vertex v of G, there is at most one spanning tree T_i such that v is not a leaf in T_i.*

Apply Theorem 1 to chordal rings, we can easily derive the following property.

Corollary 1. *Two trees T_1 and T_2 are CISTs in a chordal ring $CR(N, d)$ if and only if $E(T_1) \cap E(T_2) = \emptyset$, $|E(T_i)| = N-1$ for $i = 1, 2$, and there is no vertex v of $CR(N, d)$ such that $d_1(v) = d_2(v) = 2$, where $d_i(v)$ denotes the degree of v in T_i.*

3. Main Results

Theorem 2. *Let $N = k(d-1) + j$. For each even integer $k \geqslant 4$ and $0 \leqslant j \leqslant 4$, there are two CISTs in $CR(N, d)$.*

Proof. By $N = k(d-1) + j$ we mean that all vertices of $CR(N, d)$ are partitioned into k groups with j remanent vertices, where each group consists of $d - 1$ vertices. Let

$$V_i = \{n \mid i(d-1) \leqslant n \leqslant i(d-1) + d - 2\}$$

denote the set of vertices in the ith group of $CR(N, d)$. And, let

$$V_i^- = \{n \mid i(d-1) \leqslant n \leqslant i(d-1) + d - 3\}.$$

That is, V_i^- is the set that removes the last element from V_i. Also, for each hop $h = +1$ or $h = +d$, let $E_i(h) = \{(n, n+h) \mid n \in V_i\}$ and $E_i^-(h) = \{(n, n+h) \mid n \in V_i^-\}$ be the sets of edges incident to vertices of V_i and V_i^- with hop h, respectively.

The proof is by constructing two spanning trees, say T_B and T_R (for blue tree and red tree), in $CR(N, d)$ for each case $j = 0, 1, \ldots, 4$. Since it is easy to check up with the vertices incident to edges, we only need to provide the set of edges for describing a spanning tree.

Case $j = 0$. The constructions are

$$E(T_B) = \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+1) \right) \cup \left(\bigcup_{i=0,2,\ldots,k-4} E_i(+d) \right) \cup E^-_{k-2}(+d)$$

and

$$E(T_R) = \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+1) \right) \cup \left(\bigcup_{i=1,3,\ldots,k-3} E_i(+d) \right) \cup E^-_{k-1}(+d).$$

In the above constructions of T_B and T_R, the edge incident to a vertex $u \in V_i$ is a forward edge. Clearly, for each $i = 0, 2, \ldots, k-2$ (respectively, $i = 1, 3, \ldots, k-1$), vertices of V_i in T_B (respectively, T_R) are connected, and vertices of V_{i+1} in T_B (respectively, T_R) are connected through a path induced by V_i. Moreover, there is only one edge $(i(d-1)+d-2, (i+2)(d-1))$ with hop $+d$ that connects the two groups of vertices V_i and V_{i+2} except for $i = k-1$ and $i = k-2$. Indeed, for $(N-1) \in V_{k-1}$, we have $(N-1, N-1+d) \notin E(T_R)$; and for $(N-d) \in V_{k-2}$, we have $(N-d, 0) \notin E(T_B)$. This assures that T_B and T_R are connected and acyclic. Also, it is easy to verify that $|E(T_B)| = |E(T_R)| = N-1$, $E(T_B) \cap E(T_R) = \emptyset$ and there is no vertex of degree two in both T_B and T_R. Thus, by Corollary 1, T_B and T_R are two CISTs of $CR(N, d)$. For example, Figure 2 shows a chordal ring $CR(20, 6)$, where the set of solid lines indicates the edges of T_B, and the set of dashed lines indicates the edges of T_R.

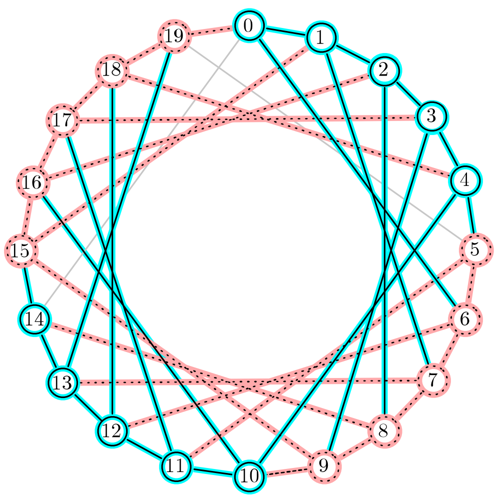

Figure 2: Two CISTs in $CR(20, 6)$.

In the remaining cases, we explicitly describe the sets of $E(T_B)$ and $E(T_R)$. Also, we omit the discussion that T_B and T_R are CISTs because all arguments are similar to case $j = 0$.

Case $j = 1$. The constructions are

$$E(T_B) = \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+1) \right) \cup \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+d) \right)$$

and

$$E(T_R) = \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+1) \right) \cup \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+d) \right)$$

For example, Figure 3 shows a chordal ring $CR(21, 6)$ and two CISTs T_B and T_R.

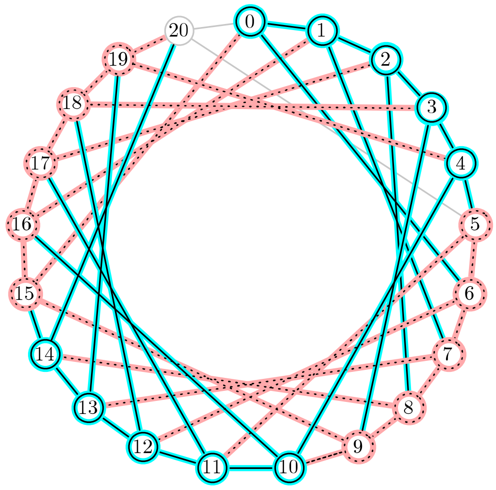

Figure 3: Two CISTs in $CR(21, 6)$.

Case $j = 2$. The constructions are

$$E(T_B) = \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+1) \right) \cup \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+d) \right) \cup \{(0, N-1)\}$$

and

$$E(T_R) = \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+1) \right) \cup \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+d) \right) \cup \{(N-2, N-2+d)\}$$

For example, Figure 4 shows a chordal ring $CR(22, 6)$ and two CISTs T_B and T_R.

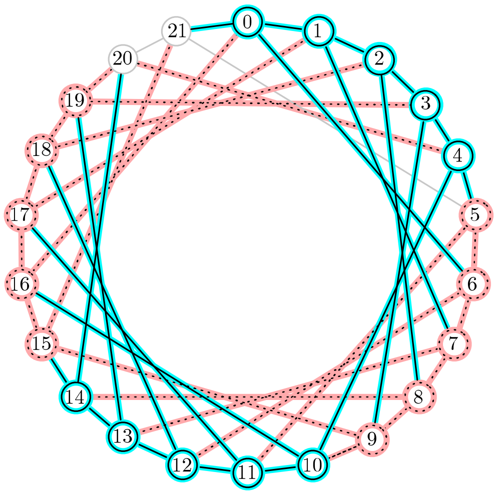

Figure 4: Two CISTs in $CR(22,6)$.

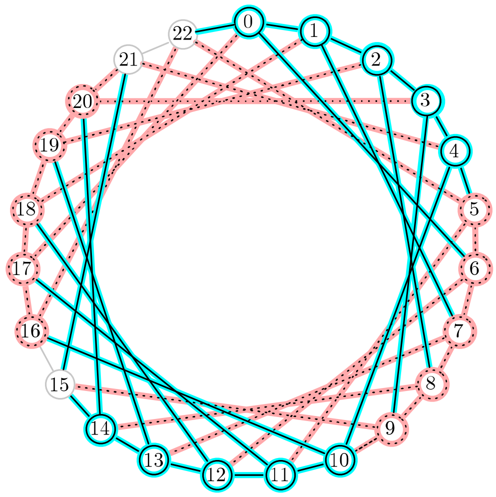

Figure 5: Two CISTs in $CR(23,6)$.

Case $j=3$. In this case, we adjust the last group as follows:

$$V_{k-1} = \{n \mid i(d-1)+1 \leqslant n \leqslant i(d-1)+d-1\}$$

and let $E_{k-1}(h) = \{(n, n+h) \mid n \in V_{k-1}\}$ for $h \in \{+1, +d\}$. Then, the constructions are

$$\begin{aligned} &E(T_B) \\ = &\left(\bigcup_{i=0,2,\ldots,k-2} E_i(+1) \right) \cup \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+d) \right) \\ &\cup \{(0, N-1), (N-2, N-2-d)\} \end{aligned}$$

and

$$\begin{aligned} &E(T_R) \\ = &\left(\bigcup_{i=1,3,\ldots,k-1} E_i(+1) \right) \cup \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+d) \right) \\ &\cup \{(N-1, N-1+d), (N-2, N-2+d)\} \end{aligned}$$

For example, Figure 5 shows a chordal ring $CR(23,6)$ and two CISTs T_B and T_R.

Case $j=4$. In this case, we adjust the last group as follows:

$$V_{k-1} = \{n \mid i(d-1)+2 \leqslant n \leqslant i(d-1)+d\}$$

and let $E_{k-1}(h) = \{(n, n+h) \mid n \in V_{k-1}\}$ for $h \in \{+1, +d\}$. Then, the constructions are

$$\begin{aligned} &E(T_B) \\ = &\left(\bigcup_{i=0,2,\ldots,k-2} E_i(+1) \right) \cup \left(\bigcup_{i=0,2,\ldots,k-2} E_i(+d) \right) \\ &\cup \{(0, N-1), (N-2, N-2-d), \\ &\quad (N-3, N-3-d)\} \end{aligned}$$

and

$$\begin{aligned} &E(T_R) \\ = &\left(\bigcup_{i=1,3,\ldots,k-1} E_i(+1) \right) \cup \left(\bigcup_{i=1,3,\ldots,k-1} E_i(+d) \right) \\ &\cup \{(N-1, N-1+d), (N-2, N-2+d), \\ &\quad (N-1-d, N-2-d)\} \end{aligned}$$

For example, Figure 6 shows a chordal ring $CR(24,6)$ and two CISTs T_B and T_R.

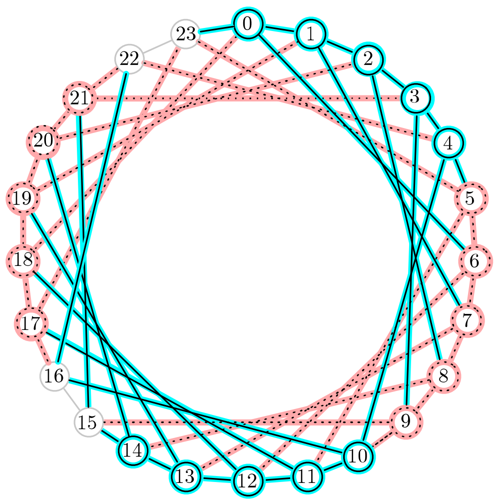

Figure 6: Two CISTs in $CR(24,6)$.

$\square$

According to Theorem 2, it is easy to compute the diameters of CISTs. Therefore, we have the following theorem.

Theorem 3. *Let $N = k(d-1)+j$. For each even integer $k \geqslant 4$ and $0 \leqslant j \leqslant 4$, the diameters of T_i for $i \in \{B, R\}$ constructed in Theorem 2 are as follows:*

$$diam(T_B) = \begin{cases} \frac{N}{2} & for\ j = 2, 4 \\ \lfloor \frac{N}{2} \rfloor + 1 & for\ j = 0, 1, 3 \end{cases}$$

and

$$diam(T_R) = \begin{cases} \lfloor \frac{N}{2} \rfloor + 1 & for\ j = 0, 1, 2, 4 \\ \lfloor \frac{N}{2} \rfloor + 2 & for\ j = 3 \end{cases}$$

4. Conclusion

In this paper, we show that there exist two CISTs in $CR(N, d)$ with $N = k(d-1)+j$ under the conditions that $k \geqslant 4$ is even and $0 \leqslant j \leqslant 4$. Since the problem of constructing two CISTs in 4-regular chordal rings has not been solved completely, we will continue this research in the future.

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